

David Henderson
Data Scientist | Machine Learning Scientist
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SUMMARY

AWS AI certified Machine Learning Scientist with 6+ years of experience in developing robust data pipelines, applying machine learning and deep learning models, performing statistical modeling and quantitative analysis, and managing large-scale, multimodal datasets. Published author with a strong record in bioinformatics, experimentation, and open-source contributions. Proven ability to collaborate with cross-functional teams to deliver high-impact analytical solutions within deadlines.

TECHNICAL SKILLS

- Machine Learning, Deep Learning (PyTorch, Keras, TensorFlow, Scikit-learn), Statistical Modeling, Quantitative Modeling, Experimentation, Forecasting
- Data Pipelines, Data Structures, Database Management, ETL, Data Transformation, Analysis, Visualization, Presentation
- Programming: Python, R, SAS, STATA, SQL, Tableau, Linux, GIS
- Amazon Web Services: Certified Machine Learning – Specialty (2025), Certified AI Practitioner (2024)

EXPERIENCE

Data Scientist Freelance

2024—Present

- Designed and implemented end-to-end data science and machine learning pipelines, including large language model (LLM) applications, for diverse projects in both business and life sciences.
- Led data collection, cleaning, integration, and exploratory analysis for high-dimensional datasets, applying advanced statistical and machine learning methods to extract actionable insights.
- Developed and maintained reproducible workflows for natural language processing, topic modeling, and classification tasks using Python, R, and cloud-based tools (e.g., AWS, Jupyter Notebooks).
- Built interactive dashboards and visualizations (e.g., Streamlit, Tableau, Shiny) to communicate results and recommendations to both technical and non-technical audiences.

Software Developer

2024

Galactic Advisors

- Developed and optimized data pipelines and SQL-based processes for data transfer, preprocessing, and analysis, ensuring high-quality data structures and forecasting capabilities.
- As part of the software development and quality control teams, utilized strict version control (Git) and development processes.
- Developed protocols for quality control. Maintained and created multi-structured database features. Optimized query rules for data transfer, analyses and automatic processes including data upload, preprocessing, formatting, analysis, and report generation.
- Updated most critical, most used sections of database schema within first 30 days.
- Collaborated with diverse teams (security, software, compliance) under tight deadlines to meet project deliverables.

Research Biologist (Ph.D.)**2018—2023**

Department of Biology, Washington University in St. Louis

- Conducted machine learning and Species Distribution Modeling analyses to determine methods to utilize less precise data resulting in enhanced functionality for the enmSdmX package in R ([link](#)).
- Spearheaded projects related to large scale network analyses using new and existing tools to interpret, analyze, and visualize multivariate relationships between hosts and pathogens
- Led 3 projects making use of bioinformatic, environmental, and multiple high-dimensional data types, developing methods to measure the impact of pathogens/natural enemies on host organisms.
- Co-organized and led an Andes-Biodiversity working group in English and Spanish with collaborators from South America, Europe and the United States that set the foundation for the future projects.

Visiting Faculty**2022**

Dr. Brian Sedio Lab, University of Texas at Austin

- Found solutions to update previous bioinformatic pipelines and created original modeling approaches to define and compare chemical profiles (consisting of thousands of unique compounds) among individuals or groups.
- Gained familiarity with the manipulation and analysis of high dimensional data and implementing ML models in scalable formats.
- Communicated with faculty and technicians to understand and perform analytic tasks including: contributions to the creation of training manuals, documentation for lab protocols and procedures, and data pipelines.

Biotech Consultant**2021—2023**

The Biotechnology and Life Science Advising (BALSA) Group

- Worked as a project-based consultant, as a team member of a non-profit, student-led consulting group.
- Selected appropriate statistical tools to address specific client needs. Summarized findings in written and oral forms.
- Performed primary and secondary research including Surveys and Interviews with KPIs, Market Segmentation, Market Sizing, Competitive Analyses (SWOT), and Exit Strategies research for academic inventors, biotech innovators and investors, in a team-based environment.
- Stayed abreast of the latest developments and innovations, and stayed current on regulations and standards relevant to the project

EDUCATION**Washington University in St. Louis****2018—2023***Ph.D.*, Biology and Biomedical Sciences**Northern Illinois University - Dekalb****2014—2016***B.Sc.*, Biological Studies – *Summa Cum Laude*